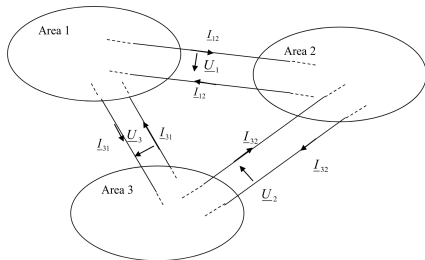


Exercises – Power grid

Exercise 1: Power-flow sign convention

Data:

$$\begin{aligned} \underline{U}_1 &= 76\angle 4^\circ \text{ kV}, & \underline{U}_2 &= 76\angle 8^\circ \text{ kV}, & \underline{U}_3 &= 76\angle -11^\circ \text{ kV}, \\ \underline{I}_{12} &= 450\angle -15^\circ \text{ A}, & \underline{I}_{31} &= 300\angle 18^\circ \text{ A}, & \underline{I}_{32} &= 500\angle -11^\circ \text{ A} \end{aligned}$$



- Which areas provide active power? How much?
- Which areas receive reactive power? How much?

$$\underline{S} = \underline{U} \cdot \underline{I}^* = \underline{U} + \underline{I}\angle(-\theta^\circ)$$

$$\begin{aligned} \underline{S}_1 &= \underline{U}_1 \underline{I}_{31}^* = 76\angle 4^\circ \cdot 450\angle 15^\circ - 76\angle -11^\circ \cdot 300\angle -18^\circ \\ &= 12.4 + j22.2 \text{ [MW]} = 12.4 \text{ [MW]} + 22.2 \text{ [MVar]} \end{aligned}$$

$$\underline{S}_2 = \underline{U}_2 \underline{I}_{32}^* - \underline{U}_1 \underline{I}_{12}^* = 3.59 \text{ [MW]} + j1.24 \text{ [MVar]}$$

$$\underline{S}_3 = \underline{U}_3 \underline{I}_{31}^* - \underline{U}_2 \underline{I}_{32}^* = -15.99 \text{ [MW]} - j23.43 \text{ [MVar]}$$

Conventionally, $S, Q > 0$: provider, $S, Q < 0$: receiver

- $\underline{S}_1$, 2: provider, 12.4 MW and 3.59 MW
- $\underline{S}_3$: receiver, 23.43 MVar

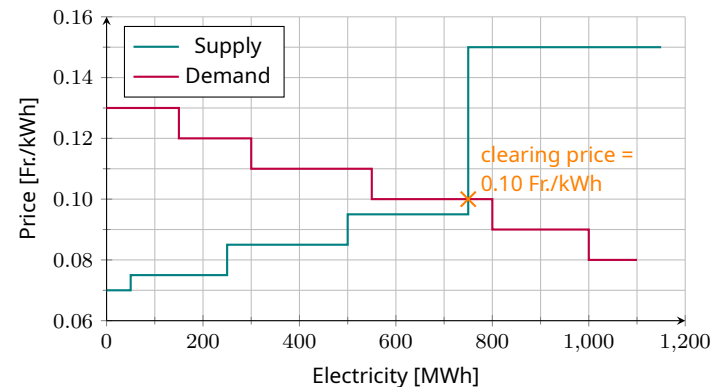
Exercise 2: Merit-order pricing

Sorted list of B1–B6 consumer offers:

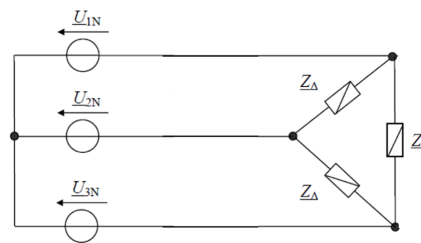
B1	B2	B3	B4	B5	B6
100 MWh 0.08 Fr./kWh	200 MWh 0.09 Fr./kWh	250 MWh 0.10 Fr./kWh	250 MWh 0.11 Fr./kWh	150 MWh 0.12 Fr./kWh	150 MWh 0.13 Fr./kWh

Sorted list of offers of electricity sellers S1–S5:

S1	S2	S3	S4	S5
50 MWh 0.07 Fr./kWh	200 MWh 0.075 Fr./kWh	250 MWh 0.085 Fr./kWh	250 MWh 0.095 Fr./kWh	400 MWh 0.15 Fr./kWh



Exercise 3: Three-phase power



Data:

$$\begin{aligned} \underline{U}_{1N} &= 110\text{ V}\angle 0^\circ, & \underline{U}_{2N} &= 230\text{ V}\angle -120^\circ, & \underline{U}_{3N} &= 230\text{ V}\angle 120^\circ, \\ \underline{Z}_\Delta &= (30 + j25) \Omega \end{aligned}$$

Calculate the complex power $\underline{S}$ and the power factor λ

$$\underline{S} = \frac{\sum_i |\underline{U}_i|^2}{\underline{Z}_\Delta^*}$$

$$\underline{U}_{12} = \underline{U}_{1N} - \underline{U}_{2N} = 110\angle 0^\circ \text{ V} - 230\angle -120^\circ \text{ V} = 300.5\angle 41.52^\circ \text{ V}$$

$$\underline{U}_{23} = \underline{U}_{2N} - \underline{U}_{3N} = 230\angle -120^\circ \text{ V} - 230\angle 120^\circ \text{ V} = 398.5\angle -90^\circ \text{ V}$$

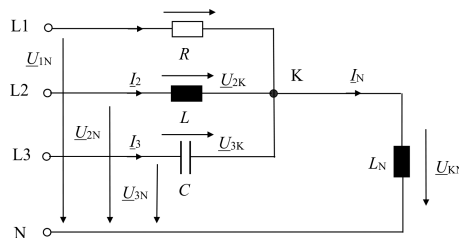
$$\underline{U}_{31} = \underline{U}_{3N} - \underline{U}_{1N} = 230\angle 120^\circ \text{ V} - 110\angle 0^\circ \text{ V} = 300.5\angle 138.48^\circ \text{ V}$$

$$\underline{S} = \frac{|\underline{U}_{12}|^2 + |\underline{U}_{23}|^2 + |\underline{U}_{31}|^2}{\underline{Z}_\Delta^*} = \frac{(300.5\text{ V})^2 + (300.5\text{ V})^2 + (398.5\text{ V})^2}{39.05\angle -39.81^\circ \Omega}$$

$$= 6.68 + j5.56 \text{ kVA} = 8.69\angle 39.81^\circ \text{ kW}$$

$$\lambda = \cos(39.81^\circ) = 0.768$$

Exercise 4: 4-wire system



Data:

$$\begin{aligned} \underline{U}_{1N} &= 220\text{ V}\angle 0^\circ, & R &= 100 \Omega, \\ \underline{U}_{2N} &= 220\text{ V}\angle -120^\circ, & L &= 160 \text{ mH}, \\ \underline{U}_{3N} &= 220\text{ V}\angle 120^\circ, & C &= 40 \mu\text{F}, \\ f &= 50 \text{ Hz}, & L_N &= 500 \text{ mH}. \end{aligned}$$

Calculate: $\underline{I}_1, \underline{I}_2, \underline{I}_3, \underline{I}_N$, and $\underline{U}_{1K}, \underline{U}_{2K}, \underline{U}_{3K}, \underline{U}_{KN}$.

$$\underline{U}_{KN} = \frac{\underline{Y}_1 \underline{U}_{1N} + \underline{Y}_2 \underline{U}_{2N} + \underline{Y}_3 \underline{U}_{3N}}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 + \underline{Y}_N}$$

$$\underline{U}_{iK} = \underline{U}_{iN} - \underline{U}_{KN} \quad ; \quad \underline{I}_i = \underline{U}_{iK} \cdot \underline{Y}_i$$

Admittances:

$$\underline{Y}_1 = \underline{Y}_R = \frac{1}{R} = 0.010\angle 0^\circ \text{ S},$$

$$\underline{Y}_2 = \underline{Y}_L = \frac{1}{\omega L} \angle -90^\circ = 0.0199\angle -90^\circ \text{ S},$$

$$\underline{Y}_3 = \underline{Y}_C = \omega C \angle 90^\circ = 0.0126\angle 90^\circ \text{ S},$$

$$\underline{Y}_N = \underline{Y}_L = \frac{1}{\omega L_N} \angle -90^\circ = 0.00637\angle -90^\circ \text{ S}.$$

Voltages:

$$\underline{U}_{KN} = \frac{\underline{Y}_1 \underline{U}_{1N} + \underline{Y}_2 \underline{U}_{2N} + \underline{Y}_3 \underline{U}_{3N}}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 + \underline{Y}_N} = 238.4\angle -137.5^\circ \text{ V},$$

$$\underline{U}_{1K} = \underline{U}_{1N} - \underline{U}_{KN} = 427.2\angle 22^\circ \text{ V},$$

$$\underline{U}_{2K} = \underline{U}_{2N} - \underline{U}_{KN} = 71.9\angle -24^\circ \text{ V},$$

$$\underline{U}_{3K} = \underline{U}_{3N} - \underline{U}_{KN} = 357.8\angle 79.4^\circ \text{ V}.$$

Currents:

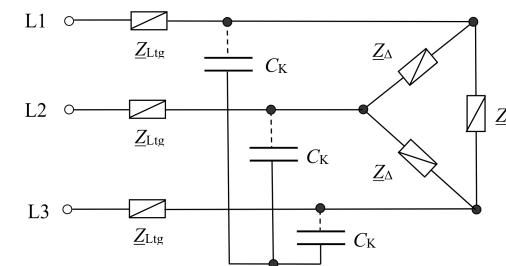
$$\underline{I}_1 = \underline{U}_{1K} \underline{Y}_1 = 4.27\angle 22^\circ \text{ A},$$

$$\underline{I}_2 = \underline{U}_{2K} \underline{Y}_2 = 1.43\angle -114^\circ \text{ A},$$

$$\underline{I}_3 = \underline{U}_{3K} \underline{Y}_3 = 4.51\angle 169.4^\circ \text{ A},$$

$$\underline{I}_N = \underline{U}_{KN} \underline{Y}_N = 1.52\angle 132.5^\circ \text{ A}.$$

Exercise 5: Three-phase systems



Data:

$$U_{LL} = 400\text{ V}/50\text{ Hz} \quad ; \quad \underline{Z}_{Ltg} = (1 + j2) \Omega \quad ; \quad \underline{Z}_\Delta = 30\angle 60^\circ$$

- Calculate the line currents $\underline{I}$ and the voltages $\underline{U}_\Delta$ across the loads $\underline{Z}_\Delta$ without the capacitor bank connected.
- Calculate the capacitance C_K of the capacitor bank for reactive current compensation of the loads $\underline{Z}_\Delta$.
- Determine the line currents $\underline{I}$ in the compensated case.

a) Without compensation:

$$\underline{I}_L = \frac{U_{ph}}{\underline{Z}_{Ltg} + \underline{Z}_Y} = \frac{U_{LL}}{\sqrt{3}} \cdot \frac{1}{\underline{Z}_{Ltg} + \frac{\underline{Z}_\Delta}{3}}$$

$$\underline{U}_\Delta = \sqrt{3} \underline{U}_Y = \sqrt{3} (\underline{I}_L \underline{Z}_Y) = \sqrt{3} \left(\underline{I}_L \cdot \frac{\underline{Z}_\Delta}{3} \right)$$

$$I_1 = \frac{400V}{\sqrt{3}} \cdot \frac{1}{(1+j2)\Omega + \frac{30\angle 60^\circ}{3}} = 18.88\angle -60.6^\circ \text{ A}$$

$$I_2 = I_1 - 120^\circ = 18.88\angle 179.4^\circ$$

$$I_3 = I_1 + 120^\circ = 18.88\angle 59.4^\circ$$

$$U_\Delta = \sqrt{3} \left(18.88\angle -60.6^\circ \cdot \frac{30\angle 60^\circ}{3} \right) = 327\angle -0.63^\circ \text{ [V]}$$

$$U_{12,\Delta} = U_\Delta + 30^\circ = 327\angle 29.7^\circ \text{ [V]}$$

$$U_{23,\Delta} = U_\Delta - 120^\circ = 327\angle -90.63^\circ \text{ [V]}$$

$$U_{31,\Delta} = U_\Delta + 120^\circ = 327\angle 149.37^\circ \text{ [V]}$$

$$Z_\Delta = \phi = 30\angle 60^\circ \Omega$$

b) Compensation capacitance:

$$Y_C = j\omega C_k \quad ; \quad Y_Y = \frac{1}{Z_Y} \quad ; \quad \text{Im}(Y_C + Y_Y) = 0$$

$$Y_Y = \frac{3}{Z_\Delta} = 0.05 - j0.0866 \text{ S} \rightarrow \text{Im}(Y_Y) = -0.0866 \text{ S}$$

$$\text{Im}(Y_C) = -Y_Y = 0.0866 \text{ S}$$

$$0.0866 = \omega C_k \Rightarrow C_k = \frac{0.0866}{2\pi \cdot 50 \text{ Hz}} = 275.8 \text{ } [\mu\text{F}]$$

c) With compensation:

$$Y_{\text{load}} = \text{Re}(Y_Y) \quad ; \quad Z_{\text{load}} = \frac{1}{Y_{\text{load}}} \quad ; \quad Z_{\text{tot}} = Z_{\text{load}} + Z_{\text{Ltg}}$$

$$Z_{\text{tot}} = 20\Omega + (1+j2)\Omega = 21 + j2\Omega$$

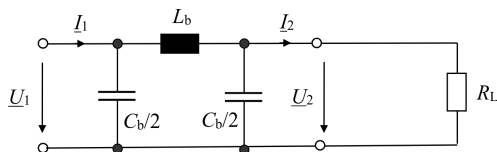
$$I_L = \frac{U_{LL}}{Z_{\text{tot}}} \rightarrow I_1 = \frac{400 \text{ [V]}}{\sqrt{3} \cdot 21 + j2\Omega} = 10.95\angle -5.44^\circ \text{ [A]}$$

$$I_2 = I_1 - 120^\circ = 10.95\angle -125.44 \text{ [A]}$$

$$I_3 = I_1 + 120^\circ = 10.95\angle 114.56 \text{ [A]}$$

Exercise 6: Surge impedance loading

Equivalent line model of a lossless line with load R_L :



Determine R_L so that the lossless line is optimal operation. **Data:**

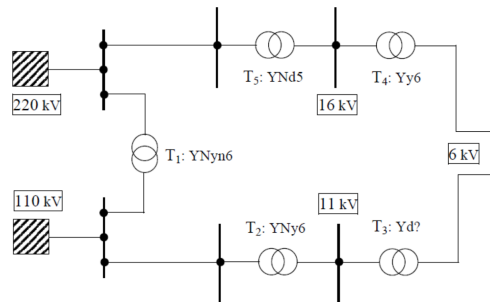
$$L'_b = 1.3 \text{ [mH/(km and phase)]} \quad ; \quad C'_b = 9.6 \text{ [nF/(km and phase)]}$$

$$l = 250 \text{ [km (length of the line)]}$$

$$Z_C = \sqrt{\frac{L'_b}{C'_b}} = \sqrt{\frac{L_b}{C_b}} \quad ; \quad L_b = L'_b \cdot l \quad ; \quad C_b = C'_b \cdot l$$

$$R_L = Z_C = \sqrt{\frac{1.3\text{E}-3}{9.6\text{E}-9}} = 368 \Omega$$

Exercise 7: Transformers 1



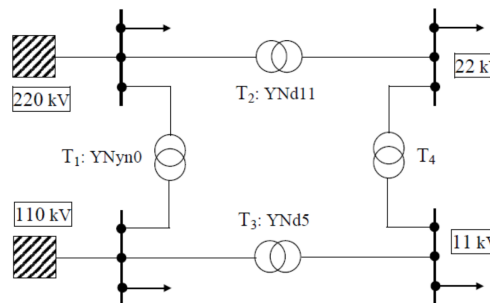
Determine the index number of transformer T_3 .

$$(T_5 + T_4) \bmod 12 = (T_3 + T_2 + T_1) \bmod 12$$

$$11 \bmod 12 = (T_3 + 12) \bmod 12$$

$$(11 - 12) \bmod 12 \equiv T_3 \rightarrow -1 \bmod 12 \equiv T_3 \rightarrow T_3 = 11$$

Exercise 8: Transformer 2



Determine a suitable transformer configuration for T_4 .

$$T_2 + T_4 = T_3 + T_2 \Rightarrow 11 + T_4 = 5$$

$$11 + T_4 \equiv 5 \bmod 12 \rightarrow T_4 = 6$$

Suitable transformers: Dd6, Yy6, Dz6.

Exercise 9: Active and reactive power of a synchronous generator

A three-phase synchronous generator (star connected) is built as a non-salient pole machine. It operates at constant voltage and frequency.

Data:

$$S_N = 400 \text{ MVA} \quad ; \quad U_N = 20 \text{ kV} / 50 \text{ Hz} \quad ; \quad X_d = 2 \text{ p.u.}$$

$$E = 25 \text{ kV} \quad ; \quad \delta = 30^\circ$$

U_N , unless otherwise specified, is line-to-line voltage.

Determine the active and reactive power of the generator. Is the machine capacitive or inductive?

$$P = \frac{U \cdot U_p}{X_d} \sin(\vartheta) \quad ; \quad P_{3p} = 3 \cdot P$$

$$Q = \frac{U \cdot U_p}{X_d} \cos(\vartheta) - \frac{U^2}{X_d} \quad ; \quad Q_{3p} = 3 \cdot Q$$

$$P = \frac{20\text{E3 [V]} \cdot 25\text{E3 [V]}}{\sqrt{3} \cdot 2 \text{ [p.u.]}} \sin(30^\circ) = 72.2 \text{ [MW]}$$

$$P_{3p} = 3 \cdot 72.2 = 216.6 \text{ [MW]}$$

$$Q = \frac{20\text{E3 [V]} \cdot 25\text{E3 [V]}}{\sqrt{3} \cdot 2 \text{ [p.u.]}} \cos(30^\circ) - \frac{\left(\frac{20\text{E3 [V]}}{\sqrt{3}} \right)^2}{2 \text{ p.u.}} = 58.33 \text{ [MVA]}$$

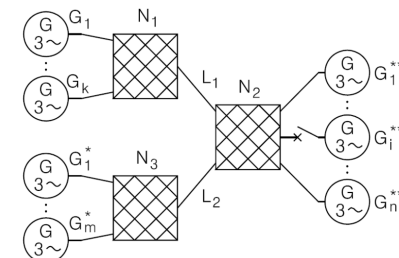
$$P_{3p} = 3 \cdot 58.33 = 175 \text{ [MVA]}$$

Exercise 10: Interconnected control areas

Three control areas N_1 - N_2 - N_3 are connected by L_1 and L_2 , initially without power exchange. A 100 MW plant fails in N_2 .

$$f_N = 50 \text{ Hz} \quad ; \quad K_{N1} = 400 \text{ MW/Hz} \quad ; \quad K_{N2} = K_{N3} = 500 \text{ MW/Hz}$$

- What frequency will occur in the control areas after the primary controllers have been activated?
- What power is then exchanged between the areas?
- What input signals ΔP_R are given to the secondary controllers of the three areas?
- When will the power exchange on L1 and L2 return to normal?



a) Frequency shift:

$$\Delta P_i = -\sum_i (K_{Ni}) \Delta f \Leftrightarrow \Delta f = \frac{\Delta P}{-(\sum_i K_{Ni})}$$

$$\Delta f = \frac{100 \text{ [MW]}}{-(400 + 500 + 500) \text{ [MW/Hz]}} = -0.0714 \text{ [Hz]}$$

b) Power exchanged:

$$\Delta P_i = -K_{Ni} \Delta f$$

$$\Delta P_1 = -400 \text{ [MW/Hz]} \cdot (-0.0714) \text{ [Hz]} = 28.56 \text{ [MW]}$$

$$\Delta P_2 = \Delta P_3 = -500 \text{ [MW/Hz]} \cdot (-0.0714) \text{ [Hz]} = 35.7 \text{ [MW]}$$

c) Secondary controllers:

$$\Delta P_{Ri} = \Delta P_i - \Delta P_{Ki}$$

$$\Delta P_{R1} = \Delta P_1 + K_{N1} \Delta f = 28.56 \text{ [MW]} - 28.56 \text{ [MW]} = 0 \text{ [MW]}$$

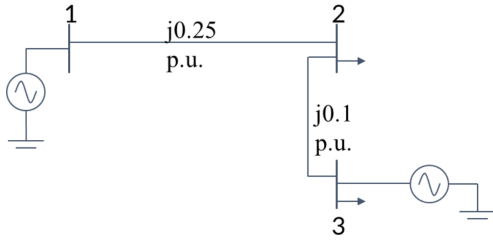
$$\Delta P_{R2} = \Delta P_3 + K_{N3} \Delta f = 35.7 \text{ [MW]} - 35.7 \text{ [MW]} = 0 \text{ [MW]}$$

$$\Delta P_{R3} = \Delta P_2 + K_{N2} \Delta f = -(\Delta P_1 + \Delta P_3) + K_{N2} \Delta f = -(28.56 + 37.5) \text{ [MW]} + 500 \cdot (-0.0714) \text{ [MW]} = -100 \text{ [MW]}$$

d) Time period: Between 5 to 15 minutes.

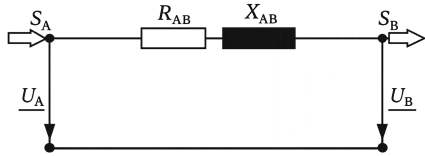
Exercise 11: Ybus

Form the Ybus of the following system:



$$\begin{cases} I_1 = \frac{1}{j0.25} (v_1 - v_2) \\ I_2 = \frac{1}{j0.1} (v_2 - v_3) + \frac{1}{j0.25} (v_2 - v_1) \\ I_3 = \frac{1}{j0.1} (v_3 - v_2) \end{cases} \quad \begin{cases} I_1 = -j4v_1 + j4v_2 \\ I_2 = -j14v_2 + j10v_3 + j4v_1 \\ I_3 = -j10v_3 + j10v_2 \end{cases}$$

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -j4 & j4 & 0 \\ j4 & -j14 & j10 \\ 0 & j10 & -j10 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

Exercise 12a: Two-node system (P_B and Q_B)Data:

$$\underline{U}_A = 400 \angle 0^\circ [\text{V}], \quad Z_{AB} = 0.1 + j0.1 [\Omega], \quad \underline{U}_B = 396.3 e^{j(-0.1774^\circ)} [\text{V}]$$

Calculate P_B and Q_B . Is the node inductive or capacitive?

$$\begin{aligned} Z_{AB} &= R_{AB} + jX_{AB} = |Z_{AB}| e^{j\theta} \\ P_B &= \frac{|U_A| |U_B|}{|Z_{AB}|} \cos(\theta + \delta) - \frac{|U_B|^2}{|Z_{AB}|} \cos(\theta) \\ Q_B &= \frac{|U_A| |U_B|}{|Z_{AB}|} \sin(\theta + \delta) - \frac{|U_B|^2}{|Z_{AB}|} \sin(\theta) \end{aligned}$$

$$Z_{AB} = 0.1414 e^{j45^\circ} [\Omega]$$

$$P_B = \frac{400 \cdot 396.3}{0.1414} \cos(45^\circ - 0.1774^\circ) - \frac{396.3^2}{0.1414} \cos(45^\circ) = 9815 [\text{W}]$$

$$Q_B = \frac{400 \cdot 396.3}{0.1414} \sin(45^\circ - 0.1774^\circ) - \frac{396.3^2}{0.1414} \sin(45^\circ) = 4908 [\text{Var}]$$

 $Q_B > 0 \rightarrow$ Node B consumes reactive power $\rightarrow$ Inductive system**Exercise 12b: Two-node system (Iterative process for U_B)**Data:

$$U_A = 400 [\text{V}]; \quad Z_{AB} = 0.1 + j0.1 [\Omega]; \quad P_B = 9815 [\text{W}]; \quad Q_B = 4908 [\text{Var}]$$

Calculate $\underline{U}_B$ (magnitude and phase angle δ).Step 1) Initialize $i = 0$ and $\underline{U}_B^{(0)} = \underline{U}_A$.

$$\text{Step 2) Calculate the current: } \underline{I}_B^{(i)} = \left\{ \frac{P_B + jQ_B}{\underline{U}_B^{(i)}} \right\}^*$$

$$\text{Step 3) Update the voltage: } \underline{U}_B^{(i+1)} = \underline{U}_A - Z_{AB} \underline{I}_B^{(i)}.$$

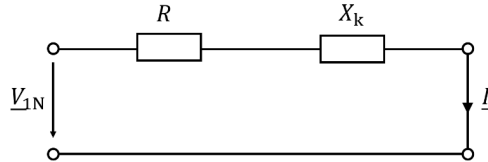
Step 4) Check convergence: $\left| \underline{U}_B^{(i+1)} \right| - \left| \underline{U}_B^{(i)} \right| < \varepsilon$; $\varepsilon = 1\text{E} - 5$.
Otherwise, set $i = i + 1$ and repeat from Step 2.

Calculator iteration:Set Ans = $\underline{U}_A$, then repeatedly enter

$$\underline{U}_A - (Z_{AB}) \frac{P_B - jQ_B}{\text{CONJ}(\text{Ans})}$$

until the result no longer changes. The final result is $\underline{U}_B$.Result:

$$\underline{U}_B = 396.28 \angle -0.18^\circ [\text{V}] = 396.28 - j1.23 [\text{V}]$$

Exercise 13: Short circuitsData:

$$V_L = 20 [\text{kV}] \quad ; \quad f = 50 [\text{Hz}] \quad ; \quad I_k = 300 [\text{A}] \quad ; \quad R = 12 [\Omega]$$

- Determine the value of the leakage reactance X_k
- Calculate the short-circuit current at time $t = 0.03$ s.
Assume that the short occurred at $t = 0$ s.

$$\begin{aligned} i(t) &= i_s(t) + i_t(t) \\ i_s(t) &= \frac{\sqrt{2}V_S}{|Z|} \sin(\omega t + \alpha - \Theta) \\ i_t(t) &= -i_s(0) \cdot e^{-(R/L)t} = \frac{\sqrt{2}V_S}{|Z|} \sin(\Theta - \alpha) \cdot e^{-(R/L)t} \\ |Z| &= \sqrt{R^2 + \omega^2 L^2} \angle \left(\Theta = \tan^{-1} \left(\frac{\omega L}{R} \right) \right) \end{aligned}$$

a) Leakage reactance:

$$\begin{aligned} V_{1N} &= \frac{V_L}{\sqrt{3}} \quad ; \quad I_k = \frac{V_{1N}}{|Z|} \Rightarrow |Z| = \frac{V_L}{\sqrt{3}I_k} \\ |Z| &= \sqrt{R^2 + X_k^2} \Rightarrow X_k = \sqrt{|Z|^2 - R^2} = \sqrt{\left(\frac{V_L}{\sqrt{3}I_k} \right)^2 - R^2} \\ &= \sqrt{\left(\frac{20\text{E}3 [\text{V}]}{\sqrt{3} \cdot 300 [\text{A}]} \right)^2 - (12 [\Omega])^2} = 36.57 [\Omega] \end{aligned}$$

b) Short circuit:

$$X_k = 2\pi f K \Rightarrow K = \frac{X_k}{2\pi f} = \frac{36.57 [\Omega]}{2\pi \cdot 50 [\text{Hz}]} = 0.1164 [\text{H}]$$

$$i_t(0) = -i_s(0) e^{-(R/K)t}$$

$$i_t(0.03) = \sqrt{2} I_k \sin(\Theta) e^{-(12/0.1164) \cdot 0.03}$$

$$= \sqrt{2} I_k \sin \left(\arctan \left(\frac{X_k}{R} \right) \right) e^{-(12/0.1164) \cdot 0.03}$$

$$= \sqrt{2} \cdot 300 [\text{A}] \sin \left(\arctan \left(\frac{36.57 [\Omega]}{12 [\Omega]} \right) \right) e^{-(12/0.1164) \cdot 0.03}$$

$$= 18.29 [\text{A}]$$

$$\begin{aligned} i_s(0.03) &= \sqrt{2} I_k \sin \left(2\pi f \cdot 0.03 [\text{s}] - \arctan \left(\frac{X_k}{R} \right) \right) \\ &= \sqrt{2} \cdot 300 [\text{A}] \sin \left(2\pi \cdot 50 [\text{Hz}] \cdot 0.03 [\text{s}] - \arctan \left(\frac{0.1164 [\text{H}]}{12 [\Omega]} \right) \right) \\ &= 403.12 [\text{A}] \end{aligned}$$

$$i(t) = i_t(t) + i_s(t) = 403.12 [\text{A}] + 18.29 [\text{A}] = 421.41 [\text{A}]$$

SW 1: Introduction

Swiss grid levels

Voltage	Role
220/380 kV	transmission and international interconnection
~ 110 kV	sub-transmission, large generation/loads
10–40 kV	regional distribution
0.4 kV	local distribution and households

Choice of grid voltage

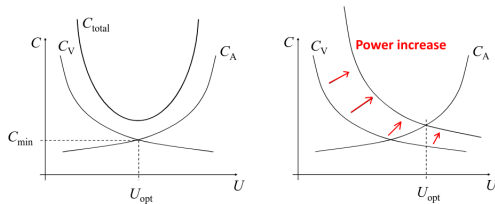
Transmission losses

$$P_{\text{trans}} = U_m I \quad ; \quad P_{\Omega} = RI^2 = R \left(\frac{P_{\text{trans}}}{U_m} \right)^2$$

$$P_{\text{trans}} = \sqrt{3} U_L I_L \cos \varphi \quad ; \quad P_{\Omega} = \frac{P_{\text{trans}}^2 R}{U_L^2 \cos^2 \varphi}$$

where: P_{trans} = transmitted active power; P_{Ω} = resistive loss; U_L = line-to-line RMS voltage.

Economic optimum



$$C_{\text{total}} = C_V + C_A$$

where: C_V = transmission-loss cost; C_A = insulation/infrastructure cost. **Rule of thumb:** economical transmission requires ~ 1 kV/km.

Reliability

N – 1 criterion: after any single component failure, the remaining system continues operating without a system-wide outage.

SAIDI: average duration of long interruptions per customer and year [min/(customer·year)].

Electricity markets

- TSO:** secure transmission-system operation; CH: Swissgrid
- Balance group:** financially responsible for deviations between scheduled and actual energy
- Futures/OTC:** contracts before delivery
- Day-ahead:** double auction for future delivery periods
- Balancing:** TSO activation close to real time

Merit order

Offers are accepted from lowest to highest marginal cost until demand is met:

$$p_{\text{market}} = \text{price of the last accepted offer}$$

SW 2: Three-Phase Systems

AC elements and phasors

$$U = \frac{\hat{U}}{\sqrt{2}} \quad ; \quad \omega = 2\pi f \quad ; \quad \underline{U} = U e^{j\varphi_u} = U \angle \varphi_u$$

$$\frac{d}{dt} \longleftrightarrow j\omega \quad ; \quad \int dt \longleftrightarrow \frac{1}{j\omega} \quad ; \quad \underline{U} = \underline{Z} \underline{I} \quad ; \quad \underline{Y} = \frac{1}{\underline{Z}}$$

	Impedance	Phase
R	$\underline{Z}_R = R$	u, i in phase
C	$\underline{Z}_C = 1/(j\omega C)$	i leads u by 90°
L	$\underline{Z}_L = j\omega L$	i lags u by 90°

Complex power

$$\underline{S} = \underline{U} \underline{I}^* = P + jQ \quad ; \quad S = UI = \sqrt{P^2 + Q^2}$$

$$P = UI \cos \varphi \quad ; \quad Q = UI \sin \varphi \quad ; \quad \lambda = \cos \varphi = \frac{P}{S}$$

where: $\varphi = \varphi_u - \varphi_i$; $Q > 0$ inductive/lagging; $Q < 0$ capacitive/leading. Sign depends on the selected load/generator current reference.

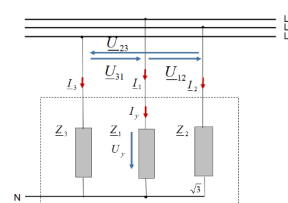
Balanced three-phase systems

$$\underline{U}_{1N} = U_{\text{ph}} \angle 0^\circ \quad ; \quad \underline{U}_{2N} = U_{\text{ph}} \angle -120^\circ \quad ; \quad \underline{U}_{3N} = U_{\text{ph}} \angle 120^\circ$$

$$\underline{U}_{12} = \underline{U}_{1N} - \underline{U}_{2N} = \sqrt{3} U_{\text{ph}} \angle 30^\circ \quad ; \quad U_{LL} = \sqrt{3} U_{\text{ph}}$$

Star and delta

Y configuration



$$U_Y = \frac{U_{LL}}{\sqrt{3}} \quad ; \quad I_Y = I_L \quad ; \quad U_{\Delta} = U_{LL} \quad ; \quad I_{\Delta} = \frac{I_L}{\sqrt{3}}$$

$$\underline{Z}_{\Delta} = 3 \underline{Z}_Y \quad ; \quad \underline{Z}_Y = \frac{\underline{Z}_{\Delta}}{3}$$

Valid for balanced operation.

Three-phase power

$$S_{3\text{ph}} = 3 \underline{U}_{\text{ph}} \underline{I}_{\text{ph}}^* = \sqrt{3} U_{LL} I_L \angle \varphi = P + jQ$$

$$S = \sqrt{3} U_{LL} I_L \quad ; \quad P = \sqrt{3} U_{LL} I_L \cos \varphi \quad ; \quad Q = \sqrt{3} U_{LL} I_L \sin \varphi$$

Unbalanced star loads

$$\underline{Y}_i = \frac{1}{\underline{Z}_i}$$

Without neutral conductor

$$\underline{U}_{KN} = \frac{\underline{Y}_1 \underline{U}_{1N} + \underline{Y}_2 \underline{U}_{2N} + \underline{Y}_3 \underline{U}_{3N}}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3}$$

With neutral impedance

$$\underline{U}_{KN} = \frac{\underline{Y}_1 \underline{U}_{1N} + \underline{Y}_2 \underline{U}_{2N} + \underline{Y}_3 \underline{U}_{3N}}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 + \underline{Y}_N} \quad ; \quad \underline{Y}_N = \frac{1}{\underline{Z}_N}$$

$$\underline{U}_{iK} = \underline{U}_{iN} - \underline{U}_{KN} \quad ; \quad \underline{I}_i = \underline{Y}_i \underline{U}_{iK} \quad ; \quad \underline{I}_N = \underline{I}_1 + \underline{I}_2 + \underline{I}_3$$

Ideal neutral: $\underline{Z}_N = 0 \Rightarrow \underline{U}_{KN} = 0$.

SW 3: Power Lines & Transformers

Stiff grid

Grid whose voltage magnitude U and frequency f remain approximately constant independently of the connected load:

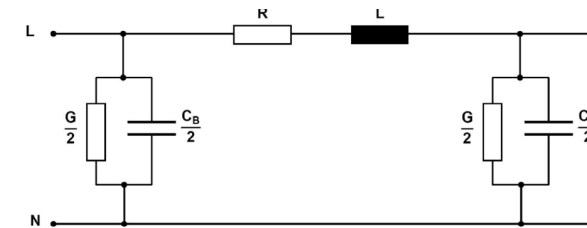
$$\Delta U \approx 0 \quad ; \quad \Delta f \approx 0$$

Equivalent to an ideal voltage source with very low internal impedance and high short-circuit power.

Equivalent line model

$$R = R' l \quad ; \quad L = L' l \quad ; \quad G = G' l \quad ; \quad C = C' l$$

$$\underline{Z} = (R' + j\omega L') l \quad ; \quad \underline{Y} = (G' + j\omega C') l$$



Length	Model	Elements
< 100 km	short	series $\underline{Z}$
100–300 km	medium	lumped Π
> 300 km	long	distributed / cascaded Π

Line parameters

$$L' = \frac{\mu_0}{2\pi} \ln \left(\frac{D}{r'} \right) \quad ; \quad C' = \frac{2\pi\epsilon\epsilon_0}{\ln(D/r')} \quad ; \quad r' \approx 0.78r$$

$$R'_{DC} = \frac{1}{\kappa A}$$

where: D = conductor spacing; r = radius; κ = conductivity. G' is often neglected.

Skin effect

AC-induced eddy currents reduce current density near the conductor centre:

$$\delta = \sqrt{\frac{2\rho}{\omega\mu}} = \sqrt{\frac{2}{\omega\mu\kappa}} \quad ; \quad R_{AC} > R_{DC}$$

where: δ = penetration depth; at 50 Hz $\delta \approx 9.38$ mm.

Surge impedance loading

$$Z_W = \sqrt{\frac{L'}{C'}}$$

For $R' = G' = 0$ and $Z_2 = Z_W$:

$$Q_1 = Q_2 = 0 \quad ; \quad P_1 = P_2 = \frac{|U_1|^2}{Z_W} = \frac{|U_2|^2}{Z_W}$$

Typical overhead line: $Z_W \approx 200\text{--}400 \Omega$.

Voltage drop along a line

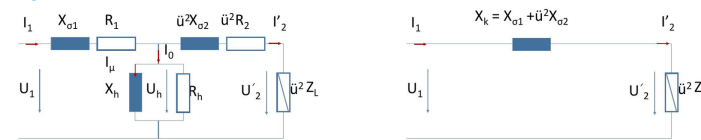
$$\underline{U}_1 = \underline{U}_2 + jX\underline{I}_2 \quad ; \quad \underline{S}_2 = P_2 + jQ_2 = \underline{U}_2 \underline{I}_2^*$$

$$\underline{U}_1 \approx \underline{U}_2 + \frac{XQ_2}{\underline{U}_2} \quad ; \quad \delta \approx \frac{XP_2}{\underline{U}_2^2}$$

Transformers**Ideal and referred quantities**

$$\underline{u} = \frac{N_1}{N_2} = \frac{\underline{U}_1}{\underline{U}_2} = \frac{\underline{I}_2}{\underline{I}_1} \quad ; \quad \underline{S}_1 = \underline{S}_2$$

$$\underline{U}'_2 = \underline{u} \underline{U}_2 \quad ; \quad \underline{I}'_2 = \frac{\underline{I}_2}{\underline{u}} \quad ; \quad \underline{Z}'_2 = \underline{u}^2 \underline{Z}_2$$

Equivalent circuit

$$R'_2 = \underline{u}^2 R_2 \quad ; \quad X'_{\sigma 2} = \underline{u}^2 X_{\sigma 2}$$

$$R_k = R_1 + R'_2 \quad ; \quad X_k = X_{\sigma 1} + X'_{\sigma 2} \quad ; \quad \underline{Z}_k = R_k + jX_k$$

Relative short-circuit voltage

$$u_k = \frac{U_{1k}}{U_{1N}} \quad ; \quad |\underline{Z}_k| = u_k \frac{U_N^2}{S_N}$$

u_k : primary-voltage fraction producing nominal current with the secondary short-circuited; typical range 4–18%.

Three-phase transformers**Connections and vector groups**

Circuit	Symbol	Character HV	Character LV	Application
Star		Y	y	High rated voltage
Delta		D	d	High rated current
Zick-Zack		Z	z	1-phase loads

HV	LV	Vector group	Application
		Yy0	Between 2 HV grids
		Yd5	Between generator and HV grid
		Yz5	Feed-in to LV grid

Vector group	Phasor diagram	Wiring diagram	
Dd0			
Yy0			
Dz0			
Dy5			
Yd5			
Yz5			

$$\underline{u} = \frac{\underline{U}_{OS}}{\underline{U}_{US}} = \frac{\underline{U}_{1U1V}}{\underline{U}_{2U2V}} = \frac{\underline{U}_{1U1N}}{\underline{U}_{2U2N}}$$

SW 4: Synchronous Generators**Three-phase generation**

A synchronous generator has a DC-excited or permanent-magnet rotor and three stator windings displaced by 120°. The rotating magnetic field induces three sinusoidal voltages.

$$U(t) = B \cdot A \cdot \omega \cdot \sin(\omega t) \quad ; \quad n = \frac{60f}{p} \quad ; \quad \omega_{\text{mech}} = \frac{\omega}{p} = \frac{2\pi f}{p} = 2\pi \frac{n}{60}$$

where: n = speed [rpm]; p = number of pole pairs. Non-salient machines are used at high speed; salient-pole machines use many poles for low-speed drives.

Non-salient vs. salient-pole generators

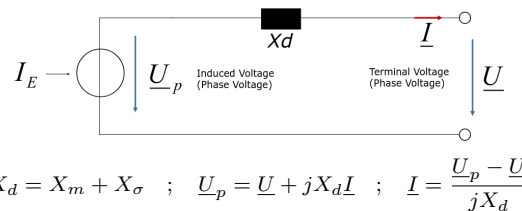
	Non-salient pole	Salient pole
Rotor	cylindrical, nearly constant air gap	projecting poles, variable air gap
Speed	high: typically 1500/3000 rpm	low: typically ≤ 1000 rpm
Pole pairs	few: usually 1–2	many
Drive	steam/gas turbine	hydro turbine
Model	$X_d = X_q$	$X_d \neq X_q$

Grid inertia

Rotating synchronous masses release kinetic energy after a power imbalance, slowing the initial frequency change. Converter-based generation requires synthetic/virtual inertia for the same function.

Equivalent circuit and torque angle

For stationary balanced operation of a non-salient generator, neglecting stator resistance:



where: $\underline{U}_p$ = induced phase voltage; $\underline{U}$ = terminal phase voltage; X_d = synchronous reactance.

$$\vartheta = \varphi_{U_p} - \varphi_U$$

$\vartheta > 0$: generator operation; $\vartheta < 0$: motor operation; $\vartheta = 0$: no active power.

Active and reactive power

With $\underline{U} = U \angle 0^\circ$ and $\underline{U}_p = U_p \angle \vartheta$:

$$P = \frac{UU_p}{X_d} \sin \vartheta \quad ; \quad Q = \frac{UU_p}{X_d} \cos \vartheta - \frac{U^2}{X_d}$$

$$P_{3\text{ph}} = 3P \quad ; \quad Q_{3\text{ph}} = 3Q$$

$$P_{\text{max},3\text{ph}} = \frac{3UU_p}{X_d} \quad \text{at} \quad \vartheta = 90^\circ$$

Operating modes and excitation

- Island operation:** excitation controls terminal voltage
- Grid connected:** grid fixes U and f ; mechanical input mainly controls P , excitation mainly controls Q
- Over-excited:** $U_p > U$, supplies Q (capacitive behaviour)
- Under-excited:** $U_p < U$, absorbs Q (inductive behaviour)

At $\vartheta = 0$:

$$P_{3\text{ph}} = 0 \quad ; \quad Q_{3\text{ph}} = \frac{3}{X_d} (U_p U - U^2)$$

Grid synchronization

Connection to the grid requires:

- equal frequency
- equal phase position
- equal voltage magnitude
- equal phase sequence L1–L2–L3

Per-unit system**Referenced quantities**

$$u = \frac{U}{U_N} \quad ; \quad i = \frac{I}{I_N} \quad ; \quad s = \frac{S}{S_N} \quad ; \quad z = \frac{Z}{Z_N}$$

Per-unit values are dimensionless. Choose U_N and S_N ; the remaining base values follow.

Three-phase base values

Using line-to-line U_N and total three-phase S_N :

$$I_N = \frac{S_N}{\sqrt{3} U_N}$$

$$Z_{N,Y} = \frac{U_N^2}{S_N} \quad ; \quad Z_{N,\Delta} = 3 \frac{U_N^2}{S_N}$$

$$Z = z Z_N$$

For the usual star-based network reference:

$$Z = z \frac{U_N^2}{S_N} \quad ; \quad z = Z \frac{S_N}{U_N^2}$$